

习题 2.27

引入 $w(x, t)$, 满足 $w(0, t) = 0$, $\partial_x w(l, t) = -\frac{Sp}{E}$, $\partial_{xx} w(x, t) = 0$

$$\Rightarrow w(x, t) = g_1(t)x + g_2(t)$$

其中 $g_1(t) = 0$, $g_2(t) = -\frac{Sp}{E}$, 则 $w(x, t) = -\frac{Sp}{E}x$

令 $v = u + w$

$$\begin{cases} \partial_{tt} v - a^2 \partial_{xx} v = g \\ v(0, t) = 0, \partial_x v(l, t) = 0 \\ u(x, 0) = \varphi(x) - \frac{Sp}{E}x, \partial_x u(x, 0) = 0 \end{cases}$$

习题 2.28

(1) 引入辅助函数 $w(x, t)$

$$-\partial_x w(0, t) + \alpha w(0, t) = -\mu_1(t), \quad \partial_x w(l, t) + \beta w(l, t) = -\mu_2(t), \quad \partial_{xx} w(x, t) = 0$$

$$\Leftrightarrow w(x, t) = g_1(t)x + g_2(t)$$

$$\begin{cases} g_1(t) - \alpha g_2(t) = +\mu_1(t) \\ g_1(t) + \beta(g_1(t)l + g_2(t)) = -\mu_2(t) \end{cases} \Leftrightarrow \begin{bmatrix} 1 & -\alpha \\ 1+\beta l & \beta \end{bmatrix} \begin{bmatrix} g_1(t) \\ g_2(t) \end{bmatrix} = \begin{bmatrix} \mu_1(t) \\ -\mu_2(t) \end{bmatrix}$$

$$\begin{bmatrix} g_1(t) \\ g_2(t) \end{bmatrix} = \frac{1}{\alpha\beta l + \alpha + \beta} \begin{bmatrix} \beta & \alpha \\ -(1+\beta l) & 1 \end{bmatrix} \begin{bmatrix} \mu_1(t) \\ -\mu_2(t) \end{bmatrix} = \frac{1}{\alpha\beta l + \alpha + \beta} \begin{bmatrix} \beta\mu_1 - \alpha\mu_2 \\ -[(1+\beta l)\mu_1 + \mu_2] \end{bmatrix}$$

令 $v = u + w$

$$\begin{cases} \partial_{tt} v - a^2 \partial_{xx} v = f(x, t) - \left[\frac{\beta\mu_1'' - \alpha\mu_2''}{\alpha + \beta + \alpha\beta l} x - \frac{(1+\beta l)\mu_1'' + \mu_2''}{\alpha + \beta + \alpha\beta l} \right], & 0 < x < l, t > 0 \\ -\partial_x v(0, t) + \alpha v(0, t) = 0, \quad \partial_x v(l, t) + \beta v(l, t) = 0, & t \geq 0 \\ v(x, 0) = \varphi(x) + \frac{1}{\alpha\beta l + \alpha + \beta} [(\beta\mu_1(0) - \alpha\mu_2(0))x - (1+\beta l)\mu_1(0) - \mu_2(0)], & 0 \leq x \leq l \end{cases}$$

$$\partial_x v(x, 0) = \psi(x) + \frac{1}{\alpha\beta l + \alpha + \beta} [(\beta\mu_1'(0) - \alpha\mu_2'(0))x - (1+\beta l)\mu_1'(0) - \mu_2'(0)], \quad 0 \leq x \leq l$$

(2) 引入辅助函数 $w(x, t)$

使 $-\partial_x w(0, t) = -\mu_1(t)$, $\partial_x w(l, t) = -\mu_2(t)$, $\partial_{xxx} w(x, t) = 0$

$$\Rightarrow w(x, t) = \cancel{g_1(t)x + g_2(t)} \quad g_1(t)x^2 + g_2(t)x + g_3(t), \quad \partial_x w(0, t) = \cancel{g_2(t)} \quad g_2(t) = \mu_1(t)$$

$$\cancel{\partial_x w(l, t) = g_1(t) = -\mu_2(t)}, \quad \partial_x w(l, t) = 2g_1(t)l + g_2(t) = -\mu_2(t)$$

$$g_1(t) = -\frac{\mu_1(t) + \mu_2(t)}{2}$$

且 $g_3(t) = 0$

$$\Rightarrow w(x, t) = -\frac{\mu_1(t) + \mu_2(t)}{2}x^2 + \mu_1(t)x$$

$$\text{令 } v = u + w$$

$$\begin{cases} \square v = f(x, t) + \left(-\frac{\mu_1'' + \mu_2''}{2} x^2 + \mu_1''(c) x + a^2(\mu_1 + \mu_2) \right) \\ -\partial_x v(0, t) = 0, \quad + \partial_x v(l, t) = 0 \\ v(x, 0) = \psi(x) - \frac{\mu_1(0) + \mu_2(0)}{2} x^2 + \mu_1(0) x, \quad \partial_x v(x, 0) = \psi'(x) - \frac{\mu_1'(0) + \mu_2'(0)}{2} x^2 + \mu_1'(0) x. \end{cases}$$

习题 2.2.9.

(1) 引入辅助函数 $w(x, t) = (1 - \frac{x}{l}) \times 0 + \frac{x}{l} (-B) = -\frac{Bx}{l}$, $v = u + w$

~~$\frac{\partial w}{\partial t} = -A$, $w(0, t) = 0$, $w(l, t) = -B$, $\frac{\partial w}{\partial x} = 0$~~

~~$w(x, t) = -\frac{1}{2}At^2 + g_1(x)t + g_2(x)$~~

~~$\frac{\partial w}{\partial x} = g_1'(x)t + g_2'(x) = 0 \Rightarrow g_1 = C_{11}x + C_{12}, g_2 = C_{21}x + C_{22}$~~

$$\begin{cases} \square v = A \\ v(0, t) = v(l, t) = 0 \\ v(x, 0) = \frac{\partial v}{\partial t}(x, 0) = 0 \end{cases}$$

再引入辅助函数 $z(x, t)$, $\square z = -A$, $z(0, t) = z(l, t) = 0$

不妨设 $z(x, t) = z(x)$, 所以 $z(x, t) = z(x) = -\frac{A}{2a^2}x(x-l)$

$\theta = v + z$

$$\begin{cases} \square \theta = 0 \\ \theta(0, t) = \theta(l, t) = 0 \\ \theta(x, 0) = \frac{A}{2a^2}x(x-l), \frac{\partial \theta}{\partial t}(x, 0) = 0 \end{cases}$$

$\theta(x, t) = X(x)T(t)$

$X(x)T''(t) = a^2 X''(x)T(t)$

$$\begin{cases} X''(x) + \lambda X(x) = 0 & X_k(x) = B_k \sin(\sqrt{\lambda_k} x), \lambda_k = (\frac{k\pi}{l})^2 \\ T''(t) + \lambda a^2 T(t) = 0 & T_k(t) = A_k \cos(a\sqrt{\lambda_k} t) \end{cases}$$

$\theta_k(x, t) = A_k B_k \sin(\frac{k\pi}{l} x) \cos(\frac{k\pi a}{l} t)$

$\theta(x, t) = \sum_{k=1}^{\infty} A_k B_k \sin(\frac{k\pi}{l} x) \cos(\frac{k\pi a}{l} t)$

$A_k B_k = \frac{2}{l} \int_0^l \frac{A}{2a^2} x(x-l) \sin(\frac{k\pi}{l} x) dx$

$$\begin{aligned} &= \frac{A}{2a^2} \int_0^l x(x-l) \sin(\frac{k\pi}{l} x) dx = \frac{A}{2\pi a^2} \left[\left(-\cos(\frac{k\pi}{l} x) x(x-l) \right)_0^l + \int_0^l \cos(\frac{k\pi}{l} x) \cdot (2x-l) dx \right] \\ &= \frac{2A}{k\pi a^2} \int_0^l \cos(\frac{k\pi}{l} x) \cdot (2x-l) dx = \frac{2A}{(k\pi a)^2} \left[(2x-l) \sin(\frac{k\pi}{l} x) \right]_0^l - \int_0^l \sin(\frac{k\pi}{l} x) \cdot 2 \cdot dx \\ &= 2A \cdot \left(\frac{l}{k\pi a} \right)^2 \cdot \frac{1}{k\pi} \cos(\frac{k\pi}{l} x) \Big|_0^l \\ &= \frac{2A}{k\pi} \cdot \left(\frac{l}{k\pi a} \right)^2 \cdot [(-1)^k - 1] \end{aligned}$$

$\theta(x, t) = \sum_{k=1}^{\infty} \frac{-4A}{(2k-1)\pi} \cdot \left[\frac{l}{(2k-1)\pi a} \right]^2 \sin\left(\frac{(2k-1)\pi}{l} x\right) \cos\left(\frac{(2k-1)\pi a}{l} t\right) \Rightarrow u(x, t) = -\frac{4A}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \left[\frac{l}{(2k-1)\pi a} \right]^2 \sin\left(\frac{(2k-1)\pi}{l} x\right)$

$$u(x, t) = \theta(x, t) - z(x, t) - w(x, t)$$

$$= -\frac{4A}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \left(\frac{l}{(2k-1)\pi a} \right)^2 \sin\left(\frac{(2k-1)\pi}{l} x\right) \cos\left(\frac{(2k-1)\pi a}{l} t\right) - \frac{A}{2a^2} x(x-l) + \frac{Bx}{l}$$

$$(2) \begin{cases} \partial_{tt} u - \partial_{xx} u = 0 \\ \partial_x u(0, t) = \sin\left(\frac{5\pi t}{2l}\right), u(l, t) = 0 \\ u(x, 0) = 1, \partial_t u(x, 0) = 0 \end{cases}$$

引入辅助函数 $w(x, t)$

$$\partial_{xx} w = 0, \partial_x w(0, t) = -\sin\left(\frac{5\pi t}{2l}\right), w(l, t) = 0$$

$$w = -\sin\left(\frac{5\pi t}{2l}\right)(x-l)$$

$$v = u + w$$

$$\begin{cases} \partial_{tt} v - \partial_{xx} v = \left(\frac{5\pi}{2l}\right)^2 \sin\left(\frac{5\pi t}{2l}\right)(x-l) = f(x, t) \\ \partial_x v(0, t) = v(l, t) = 0 \\ v(x, 0) = 1, \partial_t v(x, 0) = -\left(\frac{5\pi}{2l}\right)(x-l) \end{cases}$$

$$\text{设 } v(x, t) = X(x)T(t)$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases} \Rightarrow X_k(x) = \cos(\sqrt{\lambda_k} x), \lambda_k = \left[\frac{(k+\frac{1}{2})\pi}{l}\right]^2$$

$$= \cos\left(\frac{(k+\frac{1}{2})\pi}{l} x\right)$$

$$\text{设 } \omega_k = \frac{(k+\frac{1}{2})\pi}{l}, \text{ 则}$$

$$\begin{cases} T_k''(t) + \omega_k^2 T_k(t) = f_k(x, t) \\ T_k(0) = \varphi_k(x), T_k'(0) = \psi_k(x) \end{cases}$$

$$\begin{aligned} f_k(x, t) &= \frac{2}{l} \int_0^l \left(\frac{5\pi}{2l}\right)^2 \sin\left(\frac{5\pi t}{2l}\right)(x-l) \cos(\omega_k x) dx \\ &= \frac{2\omega_k^2}{\omega_k l} \sin(\omega_k t) \int_0^l -\sin(\omega_k x) dx \\ &= \frac{2}{l} \left(\frac{\omega_k}{\omega_l}\right)^2 \sin(\omega_k t) \end{aligned}$$

$$\varphi_k(x) = \frac{2}{l} \int_0^l \cos(\omega_k x) dx = \frac{2}{\omega_k l} (-1)^k$$

$$\psi_k(x) = -\frac{2}{l} \int_0^l \omega_k (x-l) \cos(\omega_k x) dx = \frac{2\omega_k}{l\omega_l^2}$$

$$T_k(t) = \left(f_k(x, t) * \sin \omega_k t\right) \cdot \frac{1}{\omega_k} + \varphi_k \cos \omega_k t + \frac{\psi_k}{\omega_k} \sin \omega_k t$$

$$= -\frac{2}{l} \cdot \left(\frac{\omega_k}{\omega_l}\right)^2 \cdot \frac{1}{\omega_k} \int_0^t \sin(\omega_k \tau) \sin(\omega_k(t-\tau)) d\tau + \frac{2(-1)^k}{\omega_k l} \cos \omega_k t + \frac{2\omega_k}{l\omega_l^2} \sin \omega_k t$$

$$\begin{aligned} \text{当 } k \neq 2 \text{ 时, } \int_0^t \sin(\omega_k \tau) \sin(\omega_k(t-\tau)) d\tau &= \frac{1}{2} \int_0^t [\cos((\omega_k + \omega_k)\tau - \omega_k t) - \cos((\omega_k - \omega_k)\tau + \omega_k t)] d\tau \\ &= \frac{1}{2} \left[\frac{1}{\omega_k + \omega_k} (\sin(\omega_k t) + \sin(\omega_k t)) - \frac{1}{\omega_k - \omega_k} (\sin(\omega_k t) - \sin(\omega_k t)) \right] \\ &= \frac{1}{2} \cdot \frac{1}{\omega_k^2 - \omega_k^2} (2\omega_k \sin(\omega_k t) - 2\omega_k \sin(\omega_k t)) = \frac{1}{\omega_k^2 - \omega_k^2} [\omega_k \sin(\omega_k t) - \omega_k \sin(\omega_k t)] \end{aligned}$$

$$\text{当 } k=2 \text{ 时, } \int_0^x \sin(\omega_2 \tau) \sin(\omega_2(t-\tau)) d\tau \\ = \frac{1}{2} \left[\frac{1}{\omega_2} \sin(\omega_2 t) - t \cos(\omega_2 t) \right]$$

$$\Rightarrow u(x,t) = \sum_{k=0}^{\infty} \left(\frac{\omega_2}{\omega_k} \right)^2 \cdot u(x,t) = \sum_{\substack{k \in \mathbb{N} \\ k \neq 2}} \left[-\frac{2}{l} \left(\frac{\omega_2}{\omega_k} \right)^2 \frac{1}{\omega_k} - \frac{1}{\omega_2^2 - \omega_k^2} (\omega_2 \sin(\omega_k t) - \omega_k \sin(\omega_2 t)) \right] \cos \omega_k x \\ + \left[-\frac{2}{l \omega_2} \left(\frac{1}{2} \left(\frac{1}{\omega_2} \sin \omega_2 t - t \cos \omega_2 t \right) \right) + \frac{2}{\omega_2 l} \cos \omega_2 t + \frac{2}{l \omega_2^2} \sin \omega_2 t \right] \cos \omega_2 x \\ = \sum_{k \in \mathbb{N} \setminus \{2\}} \left[-\frac{2}{l} \left(\frac{\omega_2}{\omega_k} \right)^2 \frac{1}{\omega_k} - \frac{1}{\omega_2^2 - \omega_k^2} (\omega_2 \sin(\omega_k t) - \omega_k \sin(\omega_2 t)) + \frac{2 \times (-1)^k}{\omega_k l} \cos \omega_k t + \frac{2 \omega_2}{l \omega_k^2} \sin \omega_k t \right] \cos \omega_k x \\ + \left[\frac{1}{\omega_2 l} (2+t) \cos \omega_2 t + \frac{1}{l \omega_2^2} \sin \omega_2 t \right] \cos \omega_2 x + (x-l) \sin(\omega_2 t)$$

$$(3) \text{ 记 } u(x,t) = X(x)T(t).$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ T''(t) + \lambda a^2 T(t) = 0 \\ X'(0) = X'(l) + hX(l) = 0 \end{cases} \Rightarrow \begin{cases} X_k(x) = \cos(\sqrt{\lambda_k} x) \\ X_k'(l) + hX_k(l) = 0 \end{cases} \Rightarrow \tan(\sqrt{\lambda_k} l) = \frac{h}{\sqrt{\lambda_k}}$$

$$T_k(t) = A_k \cos(a\sqrt{\lambda_k} t) + B_k \sin(a\sqrt{\lambda_k} t) = \cancel{B_k \sin(a\sqrt{\lambda_k} t)} A_k \cos(a\sqrt{\lambda_k} t) = \varphi_k \cos(a\sqrt{\lambda_k} t)$$

$$T_k(0) = \varphi_k, T_k'(0) = 0$$

$$\varphi_k = \frac{\int_0^l \cos(\sqrt{\lambda_k} x) \cos\left(\frac{\pi x}{l}\right) dx}{\int_0^l \cos^2(\sqrt{\lambda_k} x) dx} = \frac{\frac{1}{2} \left[\frac{-\sin(\sqrt{\lambda_k} l)}{\sqrt{\lambda_k} - \frac{\pi}{l}} + \frac{-\sin(\sqrt{\lambda_k} l)}{\sqrt{\lambda_k} + \frac{\pi}{l}} \right]}{\frac{1}{2} \int_0^l (1 + \cos(2\sqrt{\lambda_k} x)) dx} = \frac{-\frac{\sqrt{\lambda_k}}{\lambda_k - (\frac{\pi}{l})^2} \sin(\sqrt{\lambda_k} l)}{\frac{1}{2} \left[l + \frac{1}{2\sqrt{\lambda_k}} (\sin(2\sqrt{\lambda_k} l)) \right]} \\ = \frac{\frac{2\sqrt{\lambda_k} l}{\pi^2 - \lambda_k l^2} \sin(\sqrt{\lambda_k} l)}{1 + \frac{1}{\sqrt{\lambda_k}} \cdot \frac{2 \cdot \frac{h}{\sqrt{\lambda_k}}}{1 + \frac{h^2}{\lambda_k}}} = \frac{\frac{2\sqrt{\lambda_k} l}{\pi^2 - \lambda_k l^2} \sin(\sqrt{\lambda_k} l)}{1 + \frac{h}{l(\lambda_k + h^2)}}$$

$$\Rightarrow u(x,t) = \sum_{k=1}^{\infty} \frac{\frac{2\sqrt{\lambda_k} l}{\pi^2 - \lambda_k l^2} \sin(\sqrt{\lambda_k} l)}{1 + \frac{h}{l(\lambda_k + h^2)}} \cos(a\sqrt{\lambda_k} t) \cos(\sqrt{\lambda_k} x)$$

$$\text{其中 } \tan(\sqrt{\lambda_k} x) = \frac{h}{\sqrt{\lambda_k}}, k \in \mathbb{Z}^+$$

$$(4) \begin{cases} \partial_t u - a^2 \partial_{xx} u = 0 \\ u(0, t) = 0, u(l, t) = At \\ u(x, 0) = 0 \end{cases}$$

引入辅助函数 $w(x, t)$, $w(0, t) = 0$, $w(l, t) = -At$, $\partial_{xx} w = 0$

$$w(x, t) = -\frac{A}{l}xt$$

$$v = u + w$$

$$\begin{cases} \partial_t v - a^2 \partial_{xx} v = -\frac{A}{l}x \\ v(0, t) = v(l, t) = 0 \\ v(x, 0) = 0 \end{cases}$$

$$v(x, t) = X(x)T(t)$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases} \Rightarrow X_k(x) = \sin\left(\frac{k\pi}{l}x\right)$$

$$\begin{aligned} T_k'(t) + \left(\frac{k\pi a}{l}\right)^2 T_k(t) &= \frac{2}{l} \int_0^l -\frac{A}{l}x \sin\left(\frac{k\pi}{l}x\right) dx = \frac{2A}{k\pi l} \left[x \cos\left(\frac{k\pi x}{l}\right) \right]_0^l - \int_0^l \cos\left(\frac{k\pi x}{l}\right) dx \\ &= \frac{2A}{k\pi l} \left((-1)^k l - 0 \right) = \frac{2A \times (-1)^k}{-k\pi} \end{aligned}$$

$$T_k(0) = 0$$

$$\Rightarrow T_k(t) = \frac{2A \times (-1)^k}{-k\pi} \left(1 - e^{-\left(\frac{k\pi a}{l}\right)^2 t} \right) \cdot \left(\frac{k\pi a}{l}\right)^2$$

$$\Rightarrow u(x, t) = \sum_{k=1}^{\infty} \frac{2A \times (-1)^k \frac{k\pi a^2}{l}}{k\pi} \left(1 - e^{-\left(\frac{k\pi a}{l}\right)^2 t} \right) \sin\left(\frac{k\pi}{l}x\right) + \frac{A}{l}xt$$

$$(5) \begin{cases} \partial_t u - a^2 \partial_{xx} u = x(l-x) \\ u(0, t) = 0, \partial_x u(l, t) = 1 \\ u(x, 0) = \sin\frac{\pi}{2}x \end{cases}$$

引入辅助函数 $w(x, t)$, $w(0, t) = 0$, $\partial_x w(l, t) = -1$, $\partial_{xx} w = 0$

$$w(x, t) = -x$$

$$v = u + w$$

$$v(x, t) = X(x)T(t)$$

$$\begin{cases} \partial_t v - a^2 \partial_{xx} v = x(l-1-x) \\ v(0, t) = \partial_x v(l, t) = 0 \\ v(x, 0) = \sin\frac{\pi}{2}x - x \end{cases}$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases} \Rightarrow X_k(x) = \cos(\sqrt{\lambda_k}x), \lambda_k = \left[\frac{(k+\frac{1}{2})\pi}{l}\right]^2$$

$$\begin{cases} T_k'(t) + (\sqrt{\lambda_k})^2 T_k(t) = \frac{2}{l} \int_0^l x(l-1-x) \cos(\sqrt{\lambda_k}x) dx = \frac{2}{l} \int_0^l x(l-1-x) \cos\left(\frac{(k+\frac{1}{2})\pi x}{l}\right) dx \end{cases}$$

$$\begin{cases} T_k(0) = \frac{2}{l} \int_0^l \left[\sin\frac{\pi}{2}x - x \right] \cos\left(\frac{(k+\frac{1}{2})\pi x}{l}\right) dx \end{cases}$$

$$\begin{aligned}
 T_k'(t) + (a\omega_k)^2 T_k(t) &= \frac{2}{l} \int_0^l x(l-x) \cos(\omega_k x) dx \\
 &= \frac{2}{l\omega_k} \left[-l \cdot (-1)^{k+1} - \int_0^l \sin(\omega_k x) \cdot (-2x + l-1) dx \right] \\
 &= \frac{2}{l\omega_k} \left[(-1)^{k+1} l + \frac{1}{\omega_k} \left[(2x-l+1) \cos(\omega_k x) \right]_0^l - \int_0^l \cos \omega_k x \cdot 2 dx \right] \\
 &= \frac{2}{l\omega_k} \left[(-1)^{k+1} l + \frac{1}{\omega_k} \left[(-1)^k (l+1) - (l-1) - \frac{2}{\omega_k} (-1)^k \right] \right]
 \end{aligned}$$

$$\begin{aligned}
 T_k(0) &= \frac{2}{l} \int_0^l \left[\sin\left(\frac{\pi x}{l}\right) \cos x \right] \cos(\omega_k x) dx \\
 &= \frac{2}{l} \left[\frac{1}{2} \int_0^l \left[\sin\left(\left(\frac{\pi}{l} + \omega_k\right)x\right) + \sin\left(\left(\frac{\pi}{l} - \omega_k\right)x\right) \right] dx - \int_0^l x \cos \omega_k x dx \right] \\
 &= \frac{1}{l} \left[\frac{1}{\frac{\pi}{l} + \omega_k} + \frac{1}{\frac{\pi}{l} - \omega_k} - \frac{1}{\omega_k} \left((-1)^k l - \int_0^l \sin \omega_k x dx \right) \right] \\
 &= \frac{1}{l} \left[\frac{\frac{2\pi}{l}}{\left(\frac{\pi}{l}\right)^2 - \omega_k^2} - \frac{1}{\omega_k} \left[(-1)^k l + \frac{1}{\omega_k} (-1) \right] \right] \\
 &= \frac{1}{l} \left[\frac{\frac{2\pi}{l}}{\left(\frac{\pi}{l}\right)^2 - \omega_k^2} + \frac{(-1)^{k+1} l}{\omega_k} + \frac{1}{\omega_k^2} \right]
 \end{aligned}$$

$$\begin{aligned}
 T_k(t) &= \frac{2}{l a^2 \omega_k^3} \left[(-1)^{k+1} l + \frac{1}{\omega_k} \left((-1)^k (l+1) - (l-1) - \frac{2}{\omega_k} (-1)^k \right) \right] (1 - e^{-(a\omega_k)^2 t}) \\
 &\quad + \frac{1}{l} \left(\frac{2\pi l}{\pi^2 - l^2 \omega_k^2} + \frac{(-1)^{k+1} l}{\omega_k} + \frac{1}{\omega_k^2} \right) e^{-(a\omega_k)^2 t}
 \end{aligned}$$

$$\begin{aligned}
 u(x,t) &= \sum_{k=0}^{\infty} \left[\frac{2}{l a^2 \omega_k^3} \left((-1)^{k+1} l + \frac{1}{\omega_k} \left((-1)^k (l+1) - (l-1) - \frac{2}{\omega_k} (-1)^k \right) \right) (1 - e^{-(a\omega_k)^2 t}) + \frac{1}{l} \left(\frac{2\pi l}{\pi^2 - l^2 \omega_k^2} + \frac{(-1)^{k+1} l}{\omega_k} + \frac{1}{\omega_k^2} \right) e^{-(a\omega_k)^2 t} \right] \\
 &\quad \cos \omega_k x + x.
 \end{aligned}$$

$$(6) \begin{cases} \partial_t u - a^2 \partial_{xx} u + h(u - U_0) = 0 \\ \partial_x u(0, t) = \partial_x u(l, t) = 0 \\ u(x, 0) = \varphi(x) \end{cases}$$

$$u(x, t) = X(x) T(t)$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases} \Rightarrow \underline{X_k(x) = \cos\left(\frac{k\pi}{l}x\right)},$$

$$T_k'(t) + \left(\frac{k\pi a}{l}\right)^2 T_k(t) + h T_k(t) = \frac{2}{l} \int_0^l h U_0 \cos\left(\frac{k\pi}{l}x\right) dx = \begin{cases} 2h U_0, & k=0 \\ 0, & k \in \mathbb{Z}^+ \end{cases}$$

若 $\lambda = 0$, 則 $X_0(x) = 1$

若 $\lambda > 0$, 則 $X_k(x) = \cos\left(\frac{k\pi}{l}x\right), k \in \mathbb{Z}^+$

即 $X_k(x) = \cos\left(\frac{k\pi}{l}x\right), k \in \mathbb{N}$

$$T_k'(t) + \left[\left(\frac{k\pi a}{l}\right)^2 + h\right] T_k(t) = \begin{cases} 2h U_0, & k=0 \\ 0, & k \in \mathbb{Z}^+ \end{cases}$$

$$T_k(0) = \frac{2}{l} \int_0^l \varphi(\xi) \cos\left(\frac{k\pi}{l}\xi\right) d\xi, k > 0, T_0(0) = \frac{1}{l} \int_0^l \varphi(\xi) d\xi$$

$$T_0(t) = U_0 (1 - e^{-ht}) + \left(\frac{1}{l} \int_0^l \varphi(\xi) d\xi\right) e^{-ht}$$

$$T_k(t) = \left(\frac{2}{l} \int_0^l \varphi(\xi) \cos\left(\frac{k\pi}{l}\xi\right) d\xi\right) e^{-\left[\left(\frac{k\pi a}{l}\right)^2 + h\right]t}$$

$$u(x, t) = U_0 (1 - e^{-ht}) + \frac{1}{l} \left(\int_0^l \varphi(\xi) d\xi\right) e^{-ht} + \frac{2}{l} \sum_{k=1}^{\infty} \left(\int_0^l \varphi(\xi) \cos\left(\frac{k\pi}{l}\xi\right) d\xi\right) e^{-\left[\left(\frac{k\pi a}{l}\right)^2 + h\right]t} \cos\left(\frac{k\pi}{l}x\right)$$

习题 2.30

$$\text{若 } \partial_t u - a^2 \Delta u = 0, \partial_n u = 0 \Leftrightarrow \Delta u = 0 = \partial_{xx} u + \partial_{yy} u$$

$$(1) \begin{cases} u(0, y) = v_0, 0 < y < b \\ u(x, 0) = u(x, b) = 0, 0 \leq x \leq a \\ u(a, y) = 0, 0 \leq y \leq b \end{cases}$$

$$u(x, y) = X(x) Y(y), \begin{cases} X''(x) + \lambda X(x) = 0 \\ Y''(y) + \lambda Y(y) = 0 \end{cases} \Rightarrow Y_k(y) = \sin\left(\frac{k\pi}{b}y\right), \lambda_k = \left(\frac{k\pi}{b}\right)^2$$

$$\begin{cases} X_k(0) = \frac{2}{b} \int_0^b v_0 \sin\left(\frac{k\pi}{b}y\right) dy \\ \quad = \frac{2v_0}{k\pi} [1 - (-1)^k] = C_{k1} + C_{k2} \\ X_k(a) = 0 = C_{k1} e^{\frac{k\pi a}{b}} + C_{k2} e^{-\frac{k\pi a}{b}} \end{cases}$$

$$\begin{aligned} X_k(x) &= C_{k1} e^{\frac{k\pi}{b}x} + C_{k2} e^{-\frac{k\pi}{b}x} \\ &= \frac{2v_0}{k\pi} [1 - (-1)^k] \left(\frac{e^{-\frac{k\pi a}{b}}}{e^{\frac{k\pi a}{b}} - e^{-\frac{k\pi a}{b}}} e^{\frac{k\pi x}{b}} + \frac{e^{\frac{k\pi a}{b}}}{e^{\frac{k\pi a}{b}} - e^{-\frac{k\pi a}{b}}} e^{-\frac{k\pi x}{b}} \right) \end{aligned}$$

$$u(x, y) = \sum_{k=1}^{\infty} \frac{2V_0}{(2k+1)\pi} \frac{\sinh\left(\frac{(2k+1)\pi}{b}(a-x)\right)}{\sinh\left(\frac{(2k+1)\pi}{b}a\right)} \sin\left(\frac{(2k+1)\pi}{b}y\right)$$

$$= \sum_{k=0}^{\infty} \frac{2V_0}{(2k+1)\pi} \frac{\sinh\left[\frac{(2k+1)\pi}{b}(a-x)\right]}{\sinh\left(\frac{(2k+1)\pi}{b}a\right)} \sin\left(\frac{(2k+1)\pi}{b}y\right)$$

(2) $\begin{cases} \Delta u = 0 \\ \partial_x u(a, y) = \partial_y u(x, b) = 0 \\ u(0, y) = 0, u(x, 0) = 1 \end{cases}$

$u(x, y) = X(x) Y(y)$

$\begin{cases} X''(x) + \lambda X(x) = 0 \\ Y''(y) - \lambda Y(y) = 0 \end{cases} \quad \lambda > 0$

$X_k(x) = \cos\left(\frac{(k+\frac{1}{2})\pi}{a}x\right), \lambda_k = \left[\frac{(k+\frac{1}{2})\pi}{a}\right]^2 = \omega_k^2, k \in \mathbb{N}$

$Y_k(y) = C_{k1} e^{\omega_k y} + C_{k2} e^{-\omega_k y}$

$Y_k(0) = \frac{2}{a} \int_0^a \cos\left(\frac{(k+\frac{1}{2})\pi}{a}x\right) dx = \frac{2}{(k+\frac{1}{2})\pi} \times (-1)^k = C_{k1} + C_{k2} = \frac{2(-1)^k}{a\omega_k}$

$Y'_k(b) = \omega_k (C_{k1} e^{\omega_k b} - C_{k2} e^{-\omega_k b}) = 0$

$\begin{cases} C_{k1} + C_{k2} = \frac{2(-1)^k}{a\omega_k} \\ C_{k1} e^{\omega_k b} - C_{k2} e^{-\omega_k b} = 0 \end{cases} \Leftrightarrow \begin{cases} C_{k1} = \frac{2(-1)^k}{a\omega_k} \cdot \frac{e^{-\omega_k b}}{e^{\omega_k b} + e^{-\omega_k b}} \\ C_{k2} = \frac{2(-1)^k}{a\omega_k} \cdot \frac{e^{\omega_k b}}{e^{\omega_k b} + e^{-\omega_k b}} \end{cases}$

$Y_k(y) = \frac{2(-1)^k}{a\omega_k} \cdot \frac{\cosh[\omega_k(b-y)]}{\cosh(\omega_k b)}$

(2) $\begin{cases} \Delta u = 0 \\ \partial_x u(a, y) = \partial_y u(x, b) = 0 \\ \partial_x u(0, y) = 0, u(x, 0) = 1 \end{cases}$

$u(x, y) = X(x) Y(y)$

$\begin{cases} X''(x) + \lambda X(x) = 0 \\ Y''(y) - \lambda Y(y) = 0 \end{cases} \Rightarrow X_k(x) = \sin(\sqrt{\lambda_k} x), \lambda_k = \left[\frac{(k+\frac{1}{2})\pi}{a}\right]^2 = \omega_k^2 = \sin(\omega_k x)$

$Y_k(y) = C_{k1} e^{\omega_k y} + C_{k2} e^{-\omega_k y}$

$Y_k(0) = \frac{2}{a} \int_0^a \sin(\omega_k x) dx = \frac{2}{a\omega_k} = C_{k1} + C_{k2}$

$Y'_k(b) = 0 = C_{k1} e^{\omega_k b} - C_{k2} e^{-\omega_k b}$

$\Rightarrow Y_k(y) = \frac{2}{a\omega_k} \cdot \frac{1}{e^{\omega_k b} + e^{-\omega_k b}} (e^{\omega_k(y+b)} + e^{-\omega_k(y+b)}) = \frac{2}{a\omega_k} \cdot \frac{\cosh(y-b)}{\cosh(b)}$

$u(x, y) = \sum_{k=0}^{\infty} \frac{2}{[(k+\frac{1}{2})\pi]^2} \frac{\cosh(y-b)}{\cosh(b)} \sin\left(\frac{(k+\frac{1}{2})\pi}{a}x\right)$

$$(3) \begin{cases} \Delta u = 0 \\ u(a, y) = u(x, b) = 0 \\ u(0, y) = A \sin \frac{\pi y}{b} \\ u(x, 0) = B \sin \frac{\pi x}{a} \end{cases} \quad u(x, y) = X(x)Y(y)$$

引入辅助函数 $w(x, y)$, $w(x, 0) = -B \sin(\frac{\pi x}{a})$, $w(x, b) = 0$
 $\frac{\partial w}{\partial y} = 0$
 $\Rightarrow w(x, y) = \frac{B}{b} \sin(\frac{\pi x}{a})(y-b)$

$$v = u + w$$

$$\begin{cases} \Delta v = -(\frac{\pi}{a})^2 \cdot \frac{B}{b} \sin(\frac{\pi x}{a})(y-b) \\ v(a, y) = v(x, b) = 0 \\ v(0, y) = A \sin \frac{\pi y}{b} \\ v(x, 0) = 0 \end{cases}$$

$$v(x, y) = X(x)Y(y)$$

$$\begin{cases} Y''(y) + \lambda Y(y) = 0 \\ Y(0) = Y(b) = 0 \end{cases} \Rightarrow Y_k(y) = \sin(\frac{k\pi}{b}y), \lambda_k = (\frac{k\pi}{b})^2$$

$$X_k''(x) - (\frac{k\pi}{b})^2 X_k(x) = -(\frac{\pi}{a})^2 \cdot \frac{B}{b} \sin(\frac{\pi x}{a}) \cdot \frac{2}{b} \int_0^b (y-b) \sin(\frac{k\pi}{b}y) dy = \frac{2B}{k\pi} (\frac{\pi}{a})^2 \sin(\frac{\pi x}{a})$$

$$X_k(a) = 0$$

$$X_k(0) = A \cdot \frac{2}{b} \int_0^b \sin(\frac{\pi y}{b}) \sin(\frac{k\pi}{b}y) dy = \begin{cases} 0, & k \neq 1 \\ A, & k = 1 \end{cases}$$

$$X_k(x) = \frac{2B}{k\pi} (\frac{\pi}{a})^2 \int_0^x \sin(\frac{\pi}{a}\xi) \cdot 2 \sinh(\frac{k\pi}{b}(x-\xi)) d\xi + C_{k1} e^{\frac{k\pi}{b}x} + C_{k2} e^{-\frac{k\pi}{b}x}$$

$$\alpha = \frac{\pi}{a}, \beta_k = \frac{k\pi}{b}$$

$$\begin{aligned} \int_0^x \sin(\alpha\xi) \cdot \frac{e^{\beta_k(x-\xi)} - e^{-\beta_k(x-\xi)}}{2} d\xi &= \frac{1}{2} \left(e^{\beta_k x} \int_0^x e^{-\beta_k \xi} \sin(\alpha\xi) d\xi - e^{-\beta_k x} \int_0^x e^{\beta_k \xi} \sin(\alpha\xi) d\xi \right) \\ &= \frac{1}{2(\alpha^2 + \beta_k^2)} \left[\alpha(e^{\beta_k x} - e^{-\beta_k x}) - \beta_k \sin \alpha x - (\alpha(e^{-\beta_k x} - e^{\beta_k x}) + \beta_k \sin \alpha x) \right] \\ &= \frac{1}{2(\alpha^2 + \beta_k^2)} [\alpha(\sinh(\beta_k x)) - \beta_k \sin \alpha x] \end{aligned}$$

$$\Rightarrow X_k(x) = \frac{bB}{(k\pi a)^2} \cdot \frac{1}{(\frac{\pi}{a})^2 + (\frac{k\pi}{b})^2} \left(\left(\frac{\pi}{a}\right) \sinh\left(\frac{k\pi x}{b}\right) - \frac{k\pi}{b} \sin\left(\frac{\pi x}{a}\right) \right) + C_{k1} e^{\frac{k\pi}{b}x} + C_{k2} e^{-\frac{k\pi}{b}x}$$

$$X_k(a) = \frac{bB}{(k\pi a)^2} \cdot \frac{1}{(\frac{\pi}{a})^2 + (\frac{k\pi}{b})^2} \left(\frac{\pi}{a} \cdot \sinh\left(\frac{k\pi a}{b}\right) \right) + C_{k1} e^{\frac{k\pi a}{b}} + C_{k2} e^{-\frac{k\pi a}{b}} = 0$$

$$X_k(0) = C_{k1} + C_{k2} = \begin{cases} A, & k=1 \\ 0, & k>1 \end{cases}$$

当 $k=2$ 时,

$$C_{k1} = -C_{k2}$$

$$C_{k2} = -C_{k1} = \frac{bB}{2 \sinh(\frac{k\pi a}{b})} \cdot \frac{1}{(k\pi a)^2} \cdot \frac{1}{(\frac{\pi}{a})^2 + (\frac{k\pi}{b})^2} \cdot \frac{\pi}{a} \cdot \sinh(\frac{k\pi a}{b})$$

$$= \frac{bB\pi}{2a} \cdot \frac{1}{(k\pi a)^2} \cdot \frac{1}{(\frac{\pi}{a})^2 + (\frac{k\pi}{b})^2}$$

当 $k=1$ 时,

$$C_{12} = \frac{bB}{2 \sinh(\frac{\pi a}{b})} \cdot \frac{1}{(\pi a)^2 + (\frac{\pi}{a})^2 + (\frac{\pi}{b})^2} \cdot \frac{\pi}{a} \sinh(\frac{\pi a}{b}) + \frac{A e^{\frac{\pi a}{b}}}{2 \sinh(\frac{\pi a}{b})}$$

$$C_{11} = A - C_{12} = A \left(1 - \frac{e^{\frac{\pi a}{b}}}{2 \sinh(\frac{\pi a}{b})} \right) - \frac{bB\pi}{2a} \cdot \frac{1}{(\pi a)^2 + (\frac{\pi}{a})^2 + (\frac{\pi}{b})^2}$$

$$\Rightarrow u(x, y) = \sum_{k=1}^{\infty} \frac{bB}{(k\pi a)^2} \cdot \frac{1}{(\frac{\pi}{a})^2 + (\frac{k\pi}{b})^2} \left[\frac{\pi}{a} \sinh\left(\frac{k\pi x}{b}\right) - \frac{k\pi}{b} \sin\left(\frac{\pi x}{a}\right) \right] \sin\left(\frac{k\pi y}{b}\right)$$

$$- \sum_{k=2}^{\infty} \frac{bB\pi}{2a} \cdot \frac{1}{(k\pi a)^2} \cdot \frac{1}{(\frac{\pi}{a})^2 + (\frac{k\pi}{b})^2} \sinh\left(\frac{k\pi x}{b}\right) \sin\left(\frac{k\pi y}{b}\right)$$

$$+ \left[A \left(1 - \frac{e^{\frac{\pi a}{b}}}{2 \sinh(\frac{\pi a}{b})} \right) - \frac{bB\pi}{2a} \cdot \frac{1}{(\pi a)^2 + (\frac{\pi}{a})^2 + (\frac{\pi}{b})^2} \right] e^{\frac{\pi x}{b}}$$

$$+ \left[\frac{bB\pi}{2a} \cdot \frac{1}{(\pi a)^2 + (\frac{\pi}{a})^2 + (\frac{\pi}{b})^2} + A \cdot \frac{e^{\frac{\pi a}{b}}}{2 \sinh(\frac{\pi a}{b})} \right] e^{-\frac{\pi x}{b}} \left\{ \frac{\sinh(\frac{\pi a}{b})}{\sinh(\frac{\pi a}{b})} \right\}$$

$$+ \frac{B}{b} \sin\left(\frac{\pi x}{a}\right)(b-y)$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2}$$

$$= \frac{1}{r} \Phi Z \left(\frac{dR}{dr} + r \frac{d^2 R}{dr^2} \right) + \frac{1}{r^2} R Z \frac{d^2 \Phi}{d\varphi^2} + R \Phi \frac{d^2 Z}{dz^2} = 0$$

$$\Leftrightarrow \frac{1}{Rr} (R' + rR'') + \frac{1}{r^2 \Phi} \Phi'' + \frac{1}{Z} Z'' = 0$$

$$\Leftrightarrow \frac{r}{R} (R' + rR'') + r^2 \frac{Z''}{Z} = - \frac{\Phi''}{\Phi} \quad /$$

$$\Rightarrow \begin{cases} \Phi'' + \lambda \Phi = 0 \\ \frac{r^2}{R} (R' + rR'') + r^2 \frac{Z''}{Z} = \lambda \end{cases} \Rightarrow \Phi_k(\varphi) = \cos(\sqrt{\lambda_k} \varphi + \theta_k), \lambda_k = k^2$$

$$\Rightarrow \frac{r^2}{R_k} (R_k' + rR_k'') + r^2 \frac{Z_k''}{Z_k} = k^2$$

$$\begin{cases} Z_k'' + \mu_k Z_k = 0 \\ r^3 R_k'' + r^2 R_k' + (k^2 - \mu_k r^2) R_k = 0 \end{cases}$$

$$r^3 R_k'' + r^2 R_k' + (k^2 - \mu_k r^2) R_k = 0$$